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AI for Bharat Hackathon

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Team Name : **Team Commitement**

Team Leader Name : **Kris Patel**

Problem Statement : **AI for Retail, Commerce & Market Intelligence**

Brief about the Idea:

- **Cortexa.AI is an Autonomous Decision Intelligence platform that transforms business goals into dynamic, executable missions.**

Using multi-agent AI, it orchestrates workflows, aligns teams, and executes complex projects with real-time feedback and explainable decision-making.

Core Capabilities

- Signal Intelligence: Predictive workflow monitoring.
- Mission Orchestration: Autonomous AI task execution.
- Meeting Intelligence: Smart transcription and summaries.
- Agents IDE: AI agent design and governance.
- Collaboration Suite: Context-aware, real-time teamwork.

❖ How different is it from any of the other existing ideas?

- Intent-Driven Cognition: Strategically parses missions using LLMs to extract goals, tech stack, risks, and actionable insights.
- Autonomous Evolution: Automatically decomposes intent into prioritized tasks (P0–P3) and assigns them without manual intervention.
- Transparent AI: Provides explainable outputs and thinking logs to ensure decision clarity and eliminate black-box behavior.

❖ How will it be able to solve the problem?

- ✓ Unified Canvas: All tools and context in one dashboard.
- ✓ Autonomous Tasks: AI breaks goals into prioritized assignments.
- ✓ Signal Intelligence: Detects workflow issues in real-time.
- ✓ Explainable AI: Transparent reasoning and decision logs.

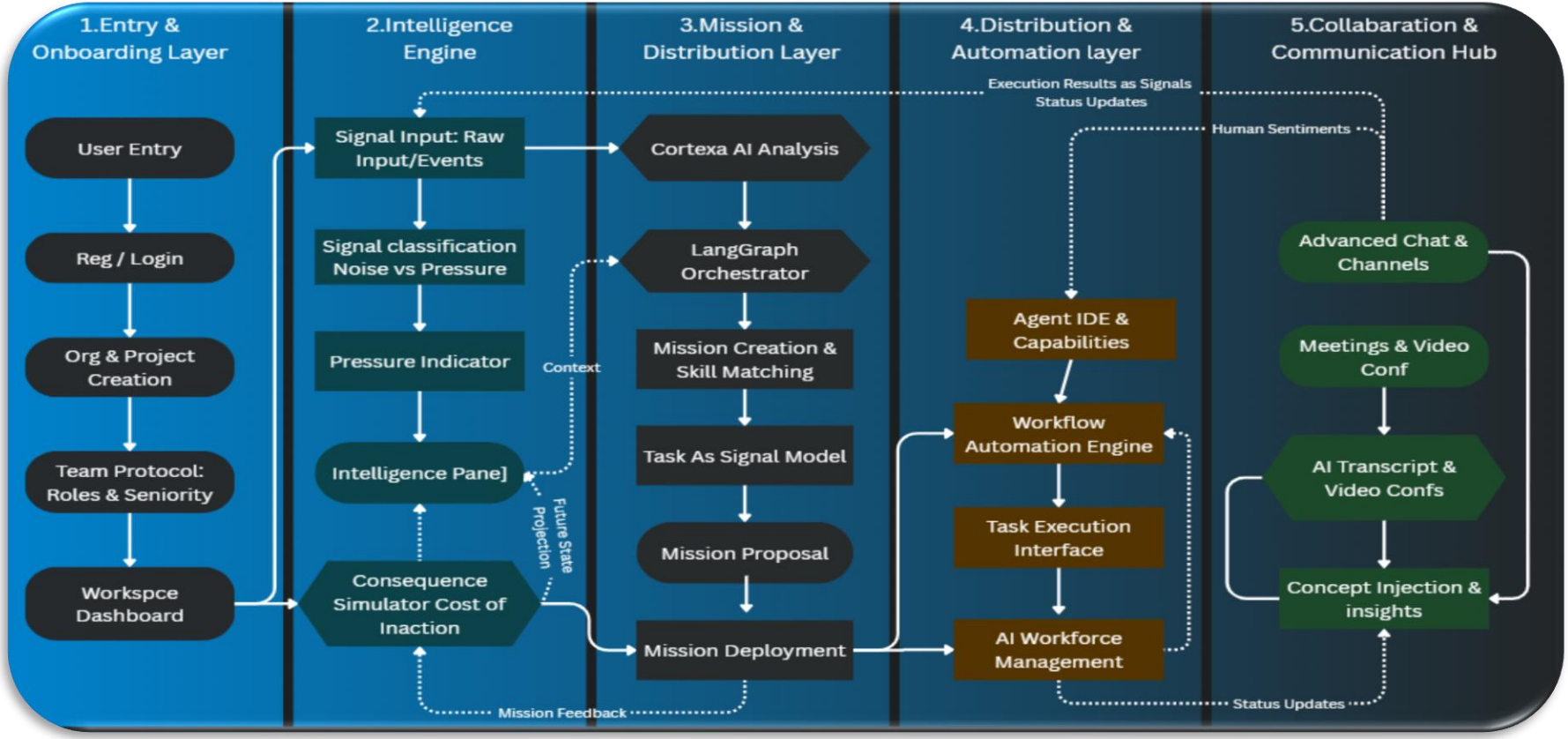
❖ USP of the proposed solution

- Zero-Touch Goal-to-Task: Autonomously breaks complex missions into prioritized task trees.
- Friction-First Intelligence: Predicts bottlenecks and detects project friction in real time.
- Radical Explainability: Streams decision traces and real-time “Thinking Logs” for full auditability.
- Hyper-Converged State: Unifies video meetings, team chat, and workflows into a single source of truth.

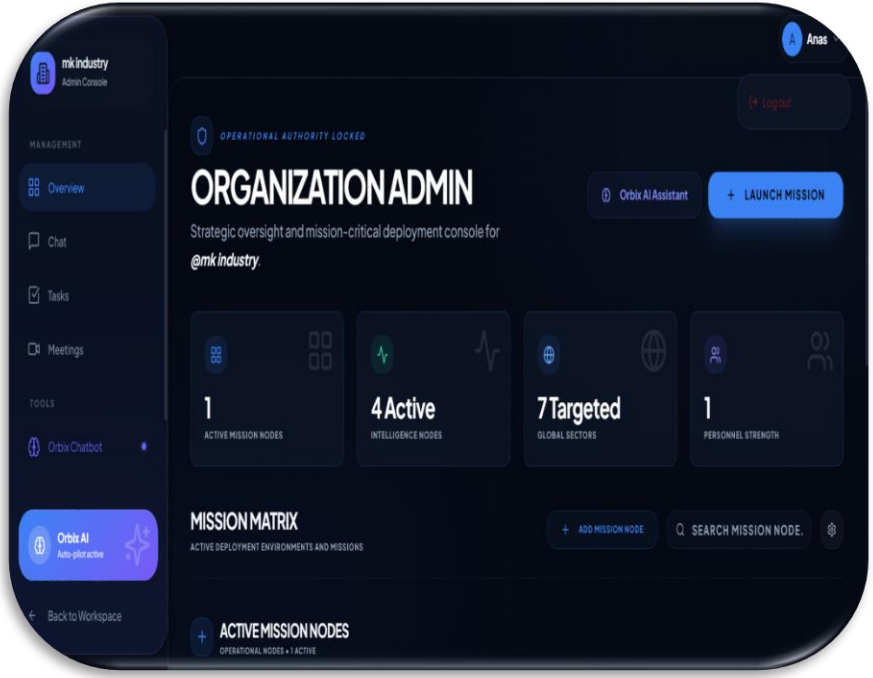
❖ List of features offered by the solution

- ✓ Intent Acquisition: Captures and semantically interprets user input to understand high-level goals and intent.
- ✓ Goal Decomposition: Converts complex missions into structured, prioritized P0–P3 task trees with dependencies and timelines.
- ✓ Knowledge Retrieval: Uses vector-based semantic search to deliver context-aware insights and mission-specific data.
- ✓ Predictive Analysis: Simulates outcomes, detects bottlenecks, and predicts workflow friction before execution.
- ✓ Decision Optimization: Matches optimal strategies with the best-suited team members using skill and availability scoring.
- ✓ Autonomous Execution: Orchestrates multi-agent workflows in real time from Draft to Completed with minimal manual oversight.
- ✓ Explainable Output: Provides transparent reasoning and decision traces for full auditability of AI actions.

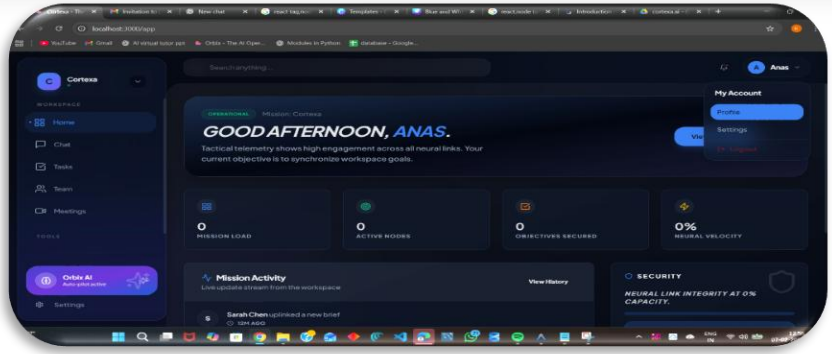
❖ Flow Diagram



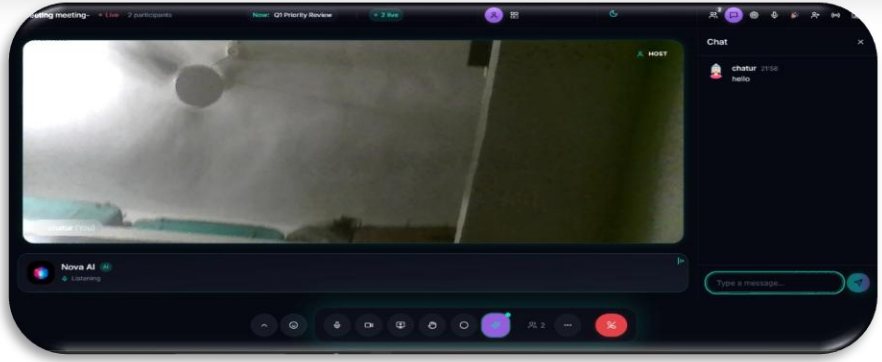
A unified gateway to orchestrate autonomous intelligence and strategic mission workflows.



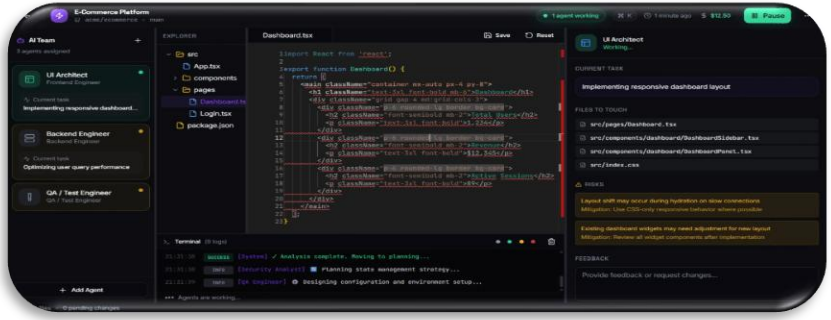
A centralized AI-powered dashboard that provides real-time operational insights, mission control, and data-driven decision support for efficient organizational management.



A unified command center for team collaboration, task management, and real-time project tracking.



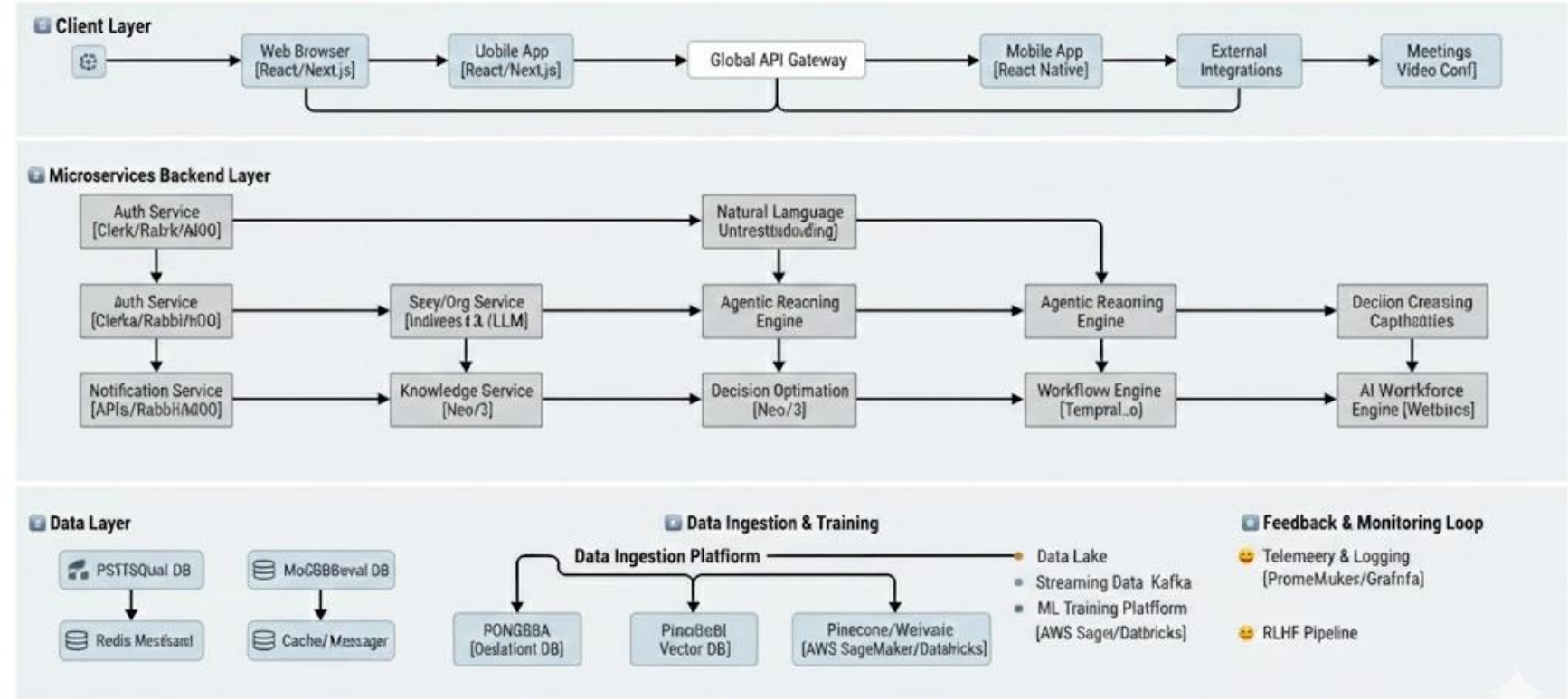
AI-Powered Meeting Interface: A smart collaboration workspace that enables real-time video meetings, live chat, and AI-assisted insights to enhance communication, decision-making, and team productivity.



AI-Assisted Development Workspace: An intelligent coding environment that streamlines software development through real-time collaboration, task automation, and AI-driven insights to improve productivity, code quality, and deployment efficiency.

AI Signal Verification Panel: A security intelligence interface that analyzes and validates detected threats, enabling teams to review, modify, and verify risk signals to ensure secure system operations and proactive vulnerability management.

Cortexa.ai Autonmous Decision Intelligerm Architecture



Cortexa.Ai operates as a Unified Neural Grid where Signals drive Pressure, Pressure shapes Missions, Missions activate Humans & Agents, and every action feeds intelligence buck into the system.

❖ Technologies to be used in the solution:

❑ The Interface Layer

- ✓ Framework: Built with React + Vite + TypeScript for a lightning-fast, enterprise-grade UI.
- ✓ Design: Styled using Tailwind CSS and shadcn/ui for an accessible, professional aesthetic.

❑ The Intelligence Core

- ✓ Reasoning Engine: LangGraph facilitates multi-step mission planning and reasoning.
- ✓ Agent Orchestration: Powered by CrewAI for seamless, multi-agent collaboration.
- ✓ Insight Engine: ProprSietary NLP Models decode intent and analyze team signals.

❑ Real-Time Infrastructure

- ✓ Backend & Data: A robust Node.js core with MongoDB for dynamic mission data management.
- ✓ Collaboration: Integrated Socket.io for live signals and Daily.co WebRTC for video-based teamwork.
- ✓ Scalability: Enterprise deployment via Vercel, Railway, and Render.

❖ Estimated implementation cost:

Service Category	Tool / Platform	Purpose	Estimated Monthly Cost
Frontend Hosting	Vercel (Free Tier)	Deploy frontend application	$0
Backend Hosting	Railway / Render (Free Tier)	Host backend services	$0
Database	MongoDB Atlas (Free Tier)	Store application data	$0
AI APIs	OpenAI GPT / Google Gemini	AI intelligence and NLP	$5 – $10 (usage-based)
Video Infrastructure	Daily.co (Free Tier)	Video meetings and transcription	$0
Authentication & Realtime	Supabase (Free Tier)	Auth and real-time services	$0
AI Frameworks	LangGraph, CrewAI	AI orchestration and agents	$0 (open source)

❖ Add as per the requirements for the hackathon:

1. AWS-Powered Infrastructure

- Elastic Compute: AWS G5/P4d GPU instances for real-time mission orchestration.
- Resilient Data: Amazon S3 and Aurora for secure, scalable mission persistence.
- Bharat Latency: AWS Local Zones (Mumbai/Delhi) for seamless Tier 2/3 city performance.

2. AWS-Powered Infrastructure

- Linguistic Inclusion: Multilingual (Hindi/Regional) support for MSME accessibility.
- Network Resilience: "Lite" agents optimized for high performance on local 4G/5G networks.
- Local Sync: Native integration with WhatsApp Business and Indian-standard ERPs.

3. Governance & Human Oversight

- Explainable AI (XAI): Transparent, auditable reasoning logs for non-technical managers.
- Compliance: Strict adherence to India's DPDP Act via local AWS data residency.
- Human-in-the-Loop: "Supervisor Mode" for manual overrides and final decision authority

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Thank You